



Sowmya Gowri <sowmya@qzense.com>

Insights on Patent Application Strategy for Our Fish Freshness Classification Process

2 messages

Sowmya Gowri <sowmya@qzense.com>
To: Srishti Batra <srishti@qzense.com>
Cc: Kartik Anagawadi <kartik@qzense.com>

Thu, May 23, 2024 at 8:00 AM

Good morning Srishti and Kartik

I hope you're doing well.

Following your request to explore patenting options for our sardine freshness classification project, I have conducted comprehensive research and gathered pertinent information. Below, I present my findings and insights regarding the patent options available, specifically focusing on the potential for applying for a utility patent for our work in India. Here are the details:

Current Workflow for Sardine Freshness Classification

Our three-step classification method is a significant innovation that could potentially qualify for a utility patent. Here's a detailed breakdown of our approach:

Step 1: Cuts and Damages Detection

- We use a YOLO V8 segmentation model to identify cuts and damages on the fish. If any are detected, the fish is classified as 'bad,' and the process stops here. This immediate classification is unique in reducing computational overhead by eliminating further steps for already bad fish.

Step 2: Size-Based Classification

- Recently integrated, this model calculates fish measurements and uses these features in a logistic regression model for binary classification. If the fish is classified as 'bad' based on size, it is not processed further. This ensures that the model identifies size-based defects early in the process.

Step 3: Softness-Based Classification

- For fish that pass the first two steps, we use a CNN model for softness classification. This model leverages transfer learning with base models like DenseNet, MobileNetV2, ResNet, or Xception. While our preprocessing functions are not novel, the combination and sequence of these models create a unique workflow.

Research on Patent Types

Process Patent vs. Utility Patent

1. Process Patent: Process patents can be applied for inventions that are methods or processes, rather than physical devices or products. In the realm of computer science, process patents can cover algorithms, computational techniques, and other step-by-step procedures that achieve a specific technical outcome.

2. Utility Patent: Utility patents protect new inventions or improvements to existing inventions in short.

Process Patent vs. Utility Patent in India

It's important to note that unlike some other jurisdictions, India doesn't have a distinct category of "utility patents." Here, all patents granted are essentially utility patents, as defined by the Patents Act, 1970. These patents protect new inventions in any field of technology, whether it's a product, a process, or a machine.

Computer Vision related patents

Obtaining a process patent for computer vision tasks is possible, but there are some key considerations:

1. Technical Advantage: The process must solve a technical problem in a new and inventive way.

2. Simply applying computer vision to an existing task might not be enough.

The focus should be on a novel and inventive technique that significantly improves upon existing computer vision methods for a particular task.

3. **Non-Obviousness:** The process shouldn't be obvious to someone skilled in computer vision. This means it should not be a routine or well-known technique in the field.

Our Invention and Patent Suitability

Based on our current three-step method for classifying sardine freshness using computer vision, I believe a **utility patent** is the most appropriate option. Here's why:

- **Focus on the Overall System:** Utility patents protect the inventive functionality of a complete system, not just individual components. Our three-step approach, combining damage/cut detection, size-based classification, and softness classification using CNNs, represents a unique system for achieving accurate freshness assessment by eliminating non-fresh fish at each step. Most existing research papers, including those I've reviewed, typically employ a single-step classification process using CNN models. This method is not commonly found in existing research or patents, making it a strong candidate for a utility patent.
- **Novelty in Combination:** While individual models or techniques might not be entirely novel, the way we've integrated them in a sequential, elimination-based process for efficient classification is likely new. This inventive combination approach strengthens our patentability argument.
- **Non-Obviousness:** Our method offers a significant advantage over existing single-step approaches by progressively eliminating classifications and reducing processing time for good quality sardines. This non-obvious improvement can be a key point in securing a patent.

Comparative Analysis of a Licensed Paper

- I reviewed several research papers and datasets, including one using Kaggle data with 5000 images. Despite high accuracy, the dataset's biased background contributed to their results, like the good images were taken in a different background and the bad images were taken in a completely different background.

- Our dataset, with consistent backgrounds and conditions for both classes, represents a more complex and realistic scenario where both the good and bad class images are taken under the same conditions and a more robust classification process.

Insights from GoMicro's Patent

- Yesterday, I reviewed a patent by GoMicro, a startup that developed a microscopic lens apparatus for mobile cameras. This apparatus functions like a microscope with curved tube, providing uniform conditions for image capture, which enhances the robustness of machine learning models trained on these images. Their patent focuses primarily on the hardware apparatus and its internal microscope, with less emphasis on AI image classification.

- The key takeaway from GoMicro's approach is the emphasis on providing uniform conditions for image capture. By using their apparatus, the accuracy of models tested on unseen data in different backgrounds remains unaffected, as the device ensures similar environmental conditions.

- Pages 17-20 and 27-30 of their patent discuss how machine learning works in general rather than specific techniques used in their software. Their focus is mainly on their unique hardware aspect, which allowed them to apply for a patent.

- They demonstrated the increased accuracy of image classification for different types of flies when using their device compared to normal images, even with a small dataset like 40 images and tested with only 9 images.

- This highlights the potential benefit of using similar devices for our project, especially for eye-based freshness classification. However, for softness classification, such devices may not be useful as softness can occur in various parts of the fish's body.

Comparison with Cited Examples

- **Agnext Patent:** The Autonomous Soil Sampler patent by Agnext appears to be a hardware invention focusing on physical components and sensor data analysis. This aligns with a process patent, as it protects the specific method of using the device for soil quality assessment. This reinforces the challenge of obtaining a process patent for purely software-based processes.
- **GoMicro Patent:** The Microscopic Lens Image Capturing Apparatus by GoMicro is another hardware-based invention. Their patent likely focuses on the design and functionality of the apparatus for capturing uniform-

condition images for training machine learning models. This again aligns with a process patent for the specific method of image acquisition.

- **Our Project vs. Cited Examples:** Our project differs from these examples because we are primarily focused on the software-based method for classifying fish freshness using computer vision.

Suggested Approach

1. Utility Patent Application:
 - Focus on the novelty of our three-step method rather than individual models or preprocessing functions.
 - Highlight the sequential elimination process and the integration of different classification steps as our unique selling points.
2. Future Enhancements:
 - Incorporate the pupil method and other advanced features in the future to strengthen our patent application.
 - Continue improving the dataset and classification accuracy to demonstrate the robustness and effectiveness of our method.
 - Consider the use of a device similar to GoMicro's lens apparatus to provide uniform conditions for image capture, further enhancing model performance and robustness.

Next Steps

- I recommend proceeding with a utility patent application emphasizing our unique process.
- This approach aligns with our current work and highlights our innovation in fish freshness classification.

References:

1. <https://www.potterclarkson.com/insights/patenting-computer-vision-inventions/>
2. <https://www.gomicro.co/sea-food/>
3. [Agnext Autonomous soil sampler patent](#)
4. [freshness classification Licensed paper](#)
5. [kaggle dataset used by the paper](#)
6. https://www.researchgate.net/publication/353117296_Fish_Freshness_Classification_Using_Combined_Deep_Learning_Model
7. <https://link.springer.com/article/10.1007/s00217-024-04493-0>
8. https://www.academia.edu/33057770/Fish_freshness_classification_based_on_image_processing_and_fuzzy_logic
9. <https://typeset.io/papers/a-vision-based-fish-processing-system-36s3xx546g>
10. https://www.researchgate.net/publication/369002421_Classification_of_Nile_Tilapia%27s_Freshness_Based_on_Eyes_and_Gills_Using_Support_Vector_Machine

I've attached the relevant research papers links and files, which used sensors for taking input and passing them to ML models, which use gills and eye images, which use feature extraction, for your reference.

Best regards,

Sowmya Maddala

2 attachments

 **go-micro-patent.pdf**
2934K

 **EI_2020_art00003_Tejaswini-Ananthanarayana.pdf**
5582K

Srishti Batra <srishti@qzense.com>
To: Sowmya Gowri <sowmya@qzense.com>
Cc: Kartik Anagawadi <kartik@qzense.com>

Thu, May 23, 2024 at 9:23 AM

Thanks Sowmya for the detailed analysis on this. I am going through it now and we will discuss this tomorrow in our meeting.

Best
srishti

[Quoted text hidden]

--

Dr. Srishti Batra
CTO | Qzense Labs Pvt. Ltd.

<https://www.qzense.com/>

<https://www.linkedin.com/in/srishtibatra/>